CSE3023 WEB APPLICATION DEVELOPMENT

SEMESTER 2 SESSION 2025/2026

BACHELOR OF COMPUTER SCIENCE (SOFTWARE ENGINEERING) WITH HONOURS

LAB REPORT 3

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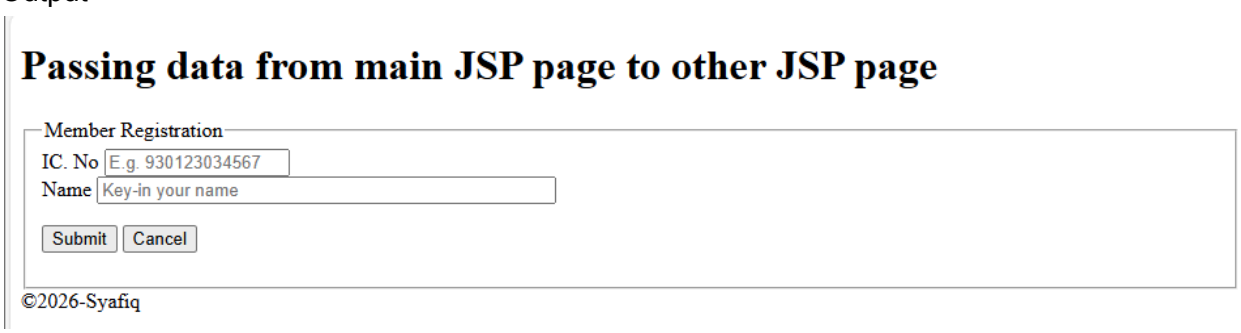
DATE OF SUBMISSION : 2 APRIL 2026

3.4 Task 1: Passing Data from Main JSP's Page to Other JSP's Page

memberRegister.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
    <head>
        <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
        <title>JSP Page</title>
    </head>
    <body>
    <body>
        <h1>Passing data from main JSP page to other JSP page</h1>
        <form id="memberFrm" action="memberProcessing.jsp" method="post"
              onsubmit="return checkICNo()">
            <fieldset>
                <legend>Member Registration</legend>
                <label for="icno">IC. No</label>
                <input type="text" id="icno" name="my_icno" size="15"
                       placeholder="E.g. 930123034567"><br/>
                <label for="icno">Name</label>
                <input type="text" id="name" name="my_name" size="45"
                       placeholder="Key-in your name"><br/>
                <p>
                    <input type="submit" id="btnSubmit" value="Submit">
                    <input type="reset" id="btnCancel" value="Cancel">
                </p>
            </fieldset>
        </form>
```

Output



memberProcessing.jsp

```jsp
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
    <head>
        <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
        <title>Lab 3-Task 1</title>
    </head>
    <body>
        <h2>Passing data from main JSP page to controller to another JSP page</h2>
    </body>
</html>
<fieldset>
    <%
    //define variables..
        String myIC = null;
        String myName = null;
        //use request.getParameter() method to retrieve data from the form in memberRegister.jsp
        myIC = request.getParameter("my_icno");
        myName = request.getParameter("my_name");
    %>
    <!-- display the output-->
    <p>Thank you for registering in this event..!</p>
    <p>This is your details:</p>
    <p>IC No : <%= myIC%></p>
    <p>Name : <%= myName%></p>
</fieldset>
```

Output

Passing data from main JSP page to controller to another JSP page

Thank you for registering in this event..!

This is your details:

IC No : 050203030177

Name : Nazmi

Reflection

1. How do you want to submit specific information from one form to the next form?
=Data is submitted using an HTML `<form>` with either `GET` or `POST` method. The data is then retrieved in the next JSP page using `request.getParameter()`.

2. What happens if the field name you use in request.getParameter("field_name") in the second page is different from the field name you defined in the first page?
=If the field name is different, the value returned will be `null`, and the data will not be displayed.

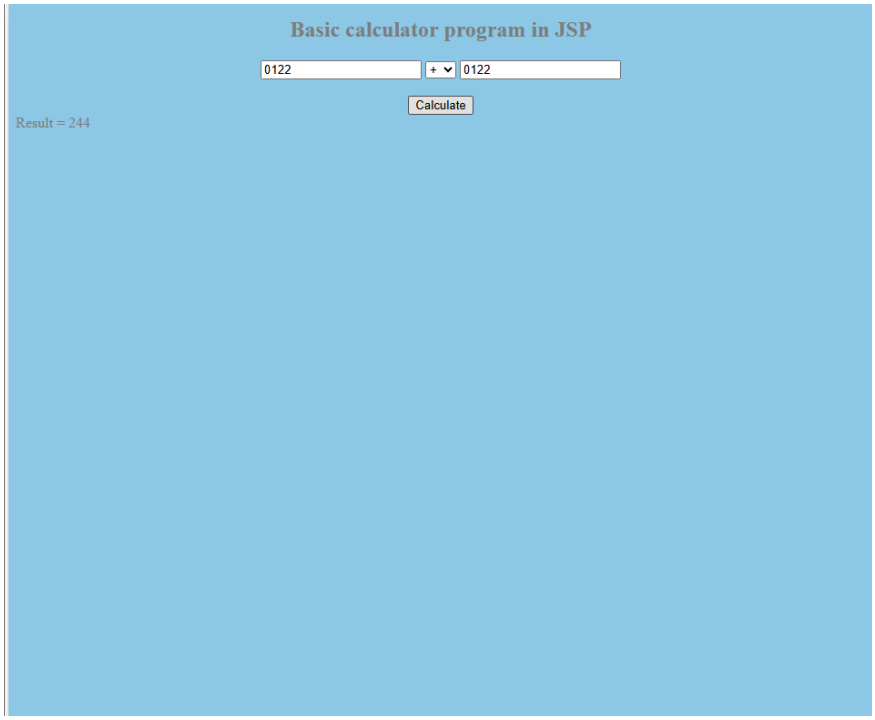
3.5 Task 2: Using Mathematics Operations in JSP

Calculator.jsp

```jsp
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
    <head>
        <meta http-equiv="Content-Type" content="text/html; charset=UTF-8"
        <title>JSP Page</title>
    </head>
    <body bgcolor="#87CEEB" text="grey">

<%
    String num1 = "0", num2 = "0";
    int result = 0;
    String op = "+";
    char opchar = op.charAt(0);
    if (request.getParameter("op") != null) {
        op = request.getParameter("op");
        opchar = op.charAt(0);
        num1 = request.getParameter("operand1");
        num2 = request.getParameter("operand2");
        switch (opchar) {
            case '0':
                result = Integer.parseInt(num1)
                        + Integer.parseInt(num2);
                break;
            case '1':
                result = Integer.parseInt(num1)
```

```jsp
            case '1':
                result = Integer.parseInt(num1)
                        - Integer.parseInt(num2);
                break;
            case '2':
                result = Integer.parseInt(num1)
                        * Integer.parseInt(num2);
                break;
            case '3':
                result = Integer.parseInt(num1)
                        / Integer.parseInt(num2);
                break;
            case '4':
                result = Integer.parseInt(num1)
                        % Integer.parseInt(num2);
                break;
        }
    }
%>
    <center>
        <h2>Basic calculator program in JSP</h2>

        <form method="get" name="f1">
            <input type="text" size="20" name="operand1" value="<%= num1 %>" />
            <select name="op" size="1">
                <option value="0">+</option>
```

```jsp
                <option value="0">+</option>
                <option value="1">-</option>
                <option value="2">*</option>
                <option value="3">/</option>
                <option value="4">%</option>
            </select>
            <input type="text" size="20" name="operand2" value="<%=num2 %>" />
            <br/><br/>
            <input type="submit" value="Calculate"/>
        </form>
 </center>
 Result = <%= result + "" %>
</body>

</html>
```



Reflection

1. How do you want to submit specific information from one form to next form?

= By using an HTML form with method="GET" or POST, and retrieving the values using request.getParameter() in the next JSP page.

2. What happens if the field name you specify in request.getParameter("field_name") in the second page is different from the field name you defined in the first page?

=The value will be null, which may cause errors such as incorrect calculations or runtime exceptions.

3.6 Task 3: Populate an Array Values into HTML's Table

```
<%--
   Document   : PopulateArray
   Created on : 14 Apr 2026, 2:57:08 PM
   Author     : MP2-4
--%>

<%@page contentType="text/html" pageEncoding="UTF-8"%>
```

```html
<!DOCTYPE html>
<html>
    <head>
        <style>
            Table {
                border-collapse: collapse;
                width: 100%;
            }

            th, td {
                text-align: left;
                padding: 8px;
            }

            tr:nth-child(even){
                background-color: #f2f2f2
            }

            th {
                background-color: orange;
                color: white;
            }
        </style>
        <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
        <title>JSP Page</title>

    </head>
    <body>


        <Table>
            <tr>
                <th> Salesman</th>
                <th> Jan</th>
                <th> Feb</th>
                <th> Mar</th>
            </tr>

            <tr>
                <td> Salesman1</td>
                <td> 2500</td>
                <td> 2100</td>
                <td> 2200</td>
            </tr>
```

```
        <tr>
            <td> Salesman2</td>
            <td> 2000</td>
            <td> 1900</td>
            <td> 2400</td>
        </tr>

        <tr>
            <td> Salesman3</td>
            <td> 1800</td>
            <td> 2200</td>
            <td> 2450</td>
        </tr>

    </Table>
  </body>
</html>
```

Salesman	Jan	Feb	Mar
Salesman1	2500	2100	2200
Salesman2	2000	1900	2400
Salesman3	1800	2200	2450

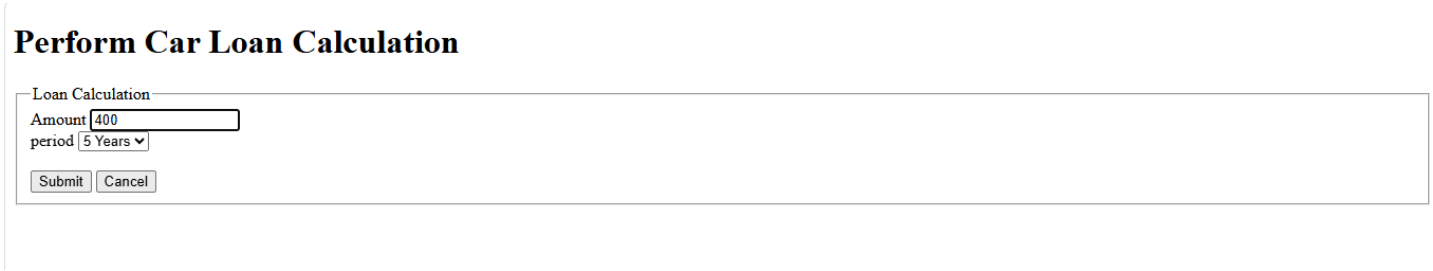
3.7 Task 4: Perform Calculation of Car Loan

calculateCarLoan.html

```
<!DOCTYPE html>

<html>
  <head>
    <title>TODO supply a title</title>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
  </head>
  <body>
    <h1>Perform Car Loan Calculation</h1>
     <form action="processCalculateCarLoan.jsp" method="post">
    <fieldset>
```

```
            <legend>Loan Calculation</legend>

            <label for="Amount">Amount</label>
            <input type="text" id="Amount" name="Amount" size="15"
                placeholder="Amount"><br/>

          <label for="period">period </label>
          <select name="period">
            <option value="3">3 Years</option>
            <option value="4">4 Years</option>
            <option value="5">5 Years</option>
            <option value="7">7 Years</option>

          </select>

          <br/><br/>
          <input type="submit" value="Submit"/>
        <input type="reset" value="Cancel"/>

      </fieldset>
      </form>
   </body>
</html>
```

Outp



processCalculateCarLoan.jsp

```
<%--
   Document   : processCalculateCarLoan.jsp
   Created on : 14 Apr 2026, 3:41:54 PM
   Author     : MP2-4
--%>

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
   <head>
```

```
        <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
        <title>Result</title>
    </head>
    <body>

        <fieldset>

    <%
    //define variables..
        String Amount = null;
        String period = null;
        //use request.getParameter() method to retrieve data from the form in memberRegister.jsp
        Amount = request.getParameter("Amount");
        period = request.getParameter("period");
    %>
    <!-- display the output-->
    <h1>Details of car loan</h1>
    <p>This is your details:</p>
    <p>LoanRequest: <%= Amount%></p>
    <p>Period of payment : <%= period %></p>

    </fieldset>
    </body>
</html>
```

Details of car loan

This is your details:

LoanRequest: 400

Period of payment : 5

Reflection

1. How you want to retrieve data from previous page?

=By using request.getParameter("fieldName") in the JSP page.

2. Where the construction of logic occur for calculating Total Loan (+ interest)

=The calculation logic is implemented in processCalculateCarLoan.jsp, where the input values are processed and computed.

3.8 Task 5: Using JSP Page Directive to Call Java API

```
<%--
   Document   : ArrayList
   Created on : 14 Apr 2026, 4:22:56 PM
   Author     : MP2-4
--%>

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%@page import="java.util.ArrayList"%>

<!DOCTYPE html>
<html>
   <head>
      <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
      <title>Use Java ArrayList</title>
   </head>
   <body>
      <h1>Use JSP Page Directive</h1>
      <br><!-- comment -->
      <%
//create ArrayList object
         ArrayList<String> studentList = new ArrayList<String>();
// store student name
         studentList.add(0, "Mohamad Azam");
         studentList.add(1, "Peter Chong");
         studentList.add(2, "Rahimah Mansor");
         studentList.add(3, "Sri Devi");
         studentList.add(4, "Ng Hue Ween");
         studentList.add(5, "S. Nagarajan");

// display the number of records
         out.println("<p>The number of records in Arraylist is "
               + studentList.size() + "</p>");
// populate list of students
         for (int i = 0; i < studentList.size(); i++)
            out.println("<p>Record " + (i + 1) + " is "
                  + studentList.get(i) + "</p>");
      %>


      <footer>%copy;2026-Syaffiq</footer>
   </body>
```

</html>

Use JSP Page Directive

The number of records in Arraylist is 6

Record 1 is Mohamad Azam

Record 2 is Peter Chong

Record 3 is Rahimah Mansor

Record 4 is Sri Devi

Record 5 is Ng Hue Ween

Record 6 is S. Nagarajan

%copy;2026-Syaffiq

Reflection

1. What you have learnt from this exercise?

= I learned how to use Java code inside JSP, work with ArrayList, and apply JSP directives to manage imports and page behavior

.

2. Write a sample syntax how you want to use java Math object in JSP?

```
<%
double result = Math.sqrt(25);
out.println(result);
%>
```

3. List and write a sample syntax for THREE (3) of JSP page directive.

```
<%@ page contentType="text/html" %>
<%@ page import="java.util.ArrayList" %>
<%@ include file="headerPage.jsp" %>
```

3.9 Task 6: Use JSP Include directive for JSP Page

footerPage.jsp

```
<%--
   Document   : footerPage
   Created on : 14 Apr 2026, 4:43:05 PM
   Author     : MP2-4
--%>

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
   <head>
      <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
      <title>JSP Page</title>
   </head>
   <style>
   .footer {
      background-color: peachpuff;
      padding: 15px;
      text-align: center;
      font-size: 24px;
      font-weight: bold;
   }
</style>

<div class="footer">
  @2026 Syafiq0
</div>

</html>
```

@2026 Syafiq0

headerPage.jsp

```
<%--
   Document   : headerPage.jsp
   Created on : 14 Apr 2026, 4:38:52 PM
   Author     : MP2-4
--%>

<%@page contentType="text/html" pageEncoding="UTF-8"%>
```

```
<!DOCTYPE html>
<html>
   <head>
      <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
      <title>JSP Page</title>
   </head>
   <style>
   .header {
      background-color: peachpuff;
      padding: 15px;
      text-align: center;
      font-size: 24px;
      font-weight: bold;
   }
</style>

<div class="header">
   ABC Sdn Bhd
</div>

</html>
```

ABC Sdn Bhd

mainPage.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>

<!DOCTYPE html>
<html>
   <head>
      <meta charset="UTF-8">
      <title>Main Page</title>
   </head>
   <body>

      <%@include file="headerPage.jsp" %>

      <h1>Using JSP Include directive</h1>
```

```
    <h4 style="color: red">
        Java Server Page (JSP) is a technology for controlling the content or appearance of
Web pages through the use of servlets, small programs that are specified in the Web page and
run on the Web server to modify the Web page before it is sent to the user who requested it
    </h4>

    <%@include file="footerPage.jsp" %>

  </body>
</html>
```

Using JSP Include directive

Java Server Page(JSP) is a technology for controlling the content or appearance of Web pages through the use of servlets, small programs that are specified in the Web page and run on the Web server to modify the Web page before it is sent to the user who requested it

ABC Sdn Bhd

Using JSP Include directive

Java Server Page (JSP) is a technology for controlling the content or appearance of Web pages through the use of servlets, small programs that are specified in the Web page and run on the Web server to modify the Web page before it is sent to the user who requested it

@2026 Syafiq0

Reflection

1. What you have learnt from this exercise?

=I learned how to use the JSP Include directive to reuse common components such as header and footer, making the web application more modular and maintainable.

2. Write a syntax how you want to include common.html file that located at a directory known as master

<%@ include file="master/common.html" %>

Exercise

1. registerClub.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<body>

<%@ include file="header.jsp" %>

<h2>Club Registration</h2>

<form action="processRegistration.jsp" method="post">
   Name:
   <input type="text" name="name" required><br><br>

   Matric No:
   <input type="text" name="matric" required><br><br>

   Club:
   <select name="club">
      <option value="Robotics Club">Robotics Club</option>
      <option value="Sports Club">Sports Club</option>
      <option value="Art Club">Art Club</option>
   </select><br><br>

   <input type="submit" value="Register">
</form>

<%@ include file="footer.jsp" %>

</body>
</html>
```

2. Processregistration.jsp

```
<%@page import="java.util.ArrayList"%>

<%@ include file="header.jsp" %>

<%
String name = request.getParameter("name");
String matric = request.getParameter("matric");
String club = request.getParameter("club");
```

```
ArrayList<String[]> members =
    (ArrayList<String[]>) application.getAttribute("members");

if (members == null) {
    members = new ArrayList<>();
}

if (name != null && matric != null && club != null) {
    members.add(new String[]{name, matric, club});
}

application.setAttribute("members", members);
%>

<h2>Registration Successful!</h2>

<p><b>Name:</b> <%= name %></p>
<p><b>Matric:</b> <%= matric %></p>
<p><b>Club:</b> <%= club %></p>

<br>
<a href="memberDirectory.jsp">Go to Member Directory</a>

<%@ include file="footer.jsp" %>
```

3. feeCalculator.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<body>

<%@ include file="header.jsp" %>

<h2>Fee Calculator</h2>

<form method="post">
    Number of Activities:
    <input type="number" name="activities" required>
    <input type="submit" value="Calculate">
</form>
```

```
<%
String act = request.getParameter("activities");

if (act != null && !act.isEmpty()) {
    int activities = Integer.parseInt(act);
    double total = activities * 10.0;
%>

<h3>Total Fee: RM <%= String.format("%.2f", total) %></h3>

<%
}
%>

<%@ include file="footer.jsp" %>

</body>
</html>
```

4. footerPage.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>

<hr>
<div style="text-align:center; font-weight:bold;">
    &copy; 2026 UMT Student Club
</div>
```

5. headerPage.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>

<style>
.header {
    background-color: peachpuff;
    padding: 15px;
    text-align: center;
    font-size: 24px;
    font-weight: bold;
}

.nav a {
    margin: 10px;
```

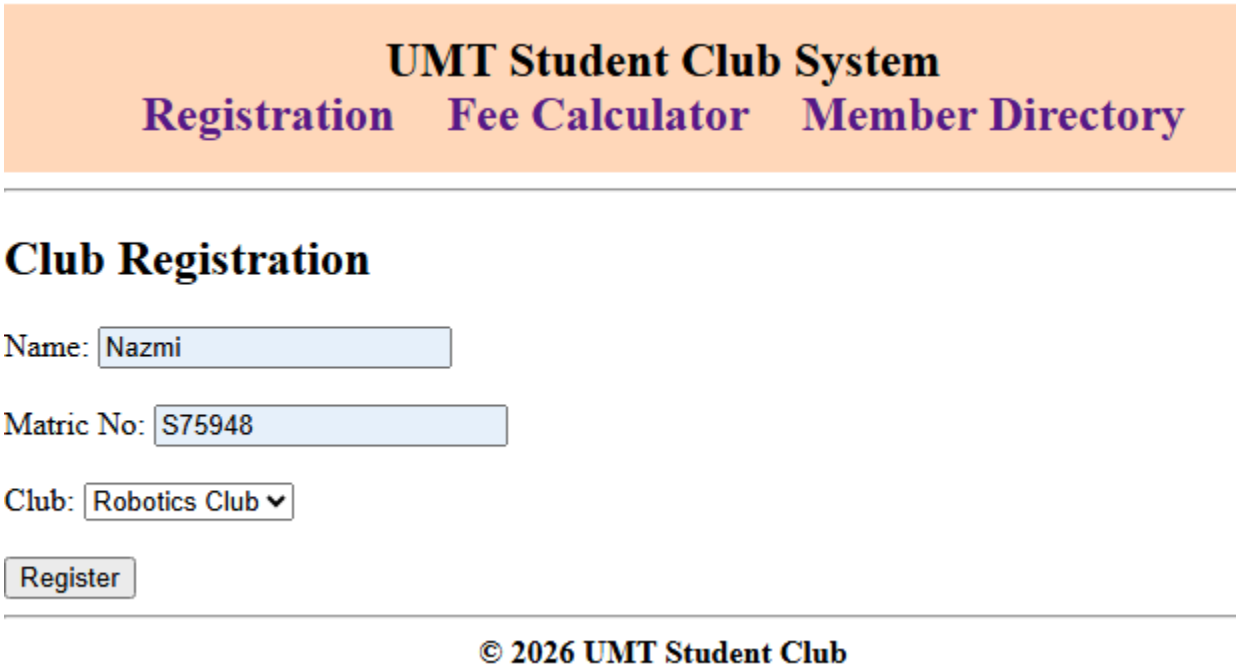
```
    text-decoration: none;
    font-weight: bold;
}
</style>

<div class="header">
    UMT Student Club System
    <div class="nav">
        <a href="registerClub.jsp">Registration</a>
        <a href="feeCalculator.jsp">Fee Calculator</a>
        <a href="memberDirectory.jsp">Member Directory</a>
    </div>
</div>

<hr>
```

6. memberDirectory.jspp

```
<%@page import="java.util.ArrayList"%>

<%@ include file="header.jsp" %>

<h2>Member Directory</h2>

<%
ArrayList<String[]> members =
    (ArrayList<String[]>) application.getAttribute("members");
%>

<table border="1" cellpadding="10">
<tr>
    <th>No</th>
    <th>Name</th>
    <th>Matric No</th>
    <th>Club</th>
</tr>

<%
if (members != null && members.size() > 0) {
    for (int i = 0; i < members.size(); i++) {
%>
<tr>
    <td><%= i + 1 %></td>
    <td><%= members.get(i)[0] %></td>
```

```
    <td><%= members.get(i)[1] %></td>
    <td><%= members.get(i)[2] %></td>
</tr>
<%
    }
} else {
%>
<tr>
    <td colspan="4">No members registered yet</td>
</tr>
<%
}
%>

</table>

<%@ include file="footer.jsp" %>
```

Output:



UMT Student Club System

Registration Fee Calculator Member Directory

Registration Successful!

Name: Nazmi

Matric: S75948

Club: Robotics Club

Go to Member Directory

© 2026 UMT Student Club

UMT Student Club System

Registration Fee Calculator Member Directory

Member Directory

No	Name	Matric No	Club
1	Nazmi	S75948	Robotics Club

© 2026 UMT Student Club

Member Directory

No	Name	Matric No	Club
1	Nazmi	S75948	Robotics Club
2	Amir	S72732	Robotics Club
3	Ali	S211	Sports Club
4	izz	S2131	Robotics Club
5	Haiqal	S1212	Robotics Club

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UMT Student Club System
Registration Fee Calculator Member Directory

Fee Calculator

Number of Activities: Calculate

Total Fee: RM 30.00

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